

PE-EC702B — Digital Image and Video Processing

Semester VII • 3L:0T:0P • 3 Credits

Module 3 Exam Notes: Color Image Processing

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Color Models

Color models provide a standardized way to specify colors. They are typically categorized as hardware-oriented (RGB, YUV) or human-oriented (HSI).

1.1 RGB Color Model

The RGB model is based on a 3D Cartesian coordinate system where the primary colors Red, Green, and Blue are at three corners. The values are typically normalized to $[0, 1]$:

- Black is at the origin $(0, 0, 0)$, while White is at $(1, 1, 1)$.
- Colors are additive: combining primaries yields secondary colors (Cyan, Magenta, Yellow).

1.2 YUV Color Model

Used widely in television transmission and video compression standards (like MPEG/H.26X). It decouples brightness (Luminance) from color (Chrominance):

- **Y (Luminance)**: Represents brightness information.
- **U and V (Chrominance)**: Represent color differences.
- Conversion from RGB to YUV:

$$Y = 0.299R + 0.587G + 0.114B \quad (1)$$

$$U = -0.14713R - 0.28886G + 0.436B \approx 0.492(B - Y) \quad (2)$$

$$V = 0.615R - 0.51499G - 0.10001B \approx 0.877(R - Y) \quad (3)$$

1.3 HSI Color Model

The HSI model decouples intensity from color-carrying information, matching human visual perception.

HSI Components

- **Hue (H)**: Describes the pure color (e.g., yellow, red, purple) represented as an angle from 0° to 360° .
- **Saturation (S)**: Represents the purity or dilution of the color with white (range $[0, 1]$).
- **Intensity (I)**: Represents the brightness or gray-level value (range $[0, 1]$).

RGB to HSI Conversion Formulas

Given $R, G, B \in [0, 1]$, the HSI components are computed as:

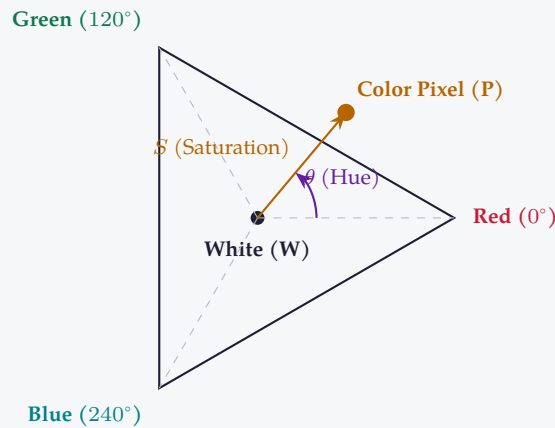
$$I = \frac{1}{3}(R + G + B) \quad (4)$$

$$S = 1 - \frac{3}{R + G + B} [\min(R, G, B)] \quad (5)$$

$$H = \begin{cases} \theta & \text{if } B \leq G \\ 360^\circ - \theta & \text{if } B > G \end{cases} \quad (6)$$

where the angle θ is:

$$\theta = \cos^{-1} \left(\frac{\frac{1}{2} [(R - G) + (R - B)]}{\sqrt{(R - G)^2 + (R - B)(G - B)}} \right) \quad (7)$$

HSI Color Model Coordinates

2. Color Transformations

Color transformations map input color pixels $\mathbf{r}$ to output pixels $\mathbf{s}$:

$$s_i = T_i(r_1, r_2, \dots, r_n) \quad \text{for } i = 1, 2, \dots, n \quad (8)$$

where n is the number of color space dimensions (e.g., $n = 3$ for RGB/HSI).

2.1 Formulations and Corrections

- **Color Complements:** Analogous to gray-scale negatives. In RGB space, the complement is obtained by subtracting the color values from the maximum:

$$s_R = 1 - r_R, \quad s_G = 1 - r_G, \quad s_B = 1 - r_B \quad (9)$$

Useful for enhancing dark details in color images.

- **Color Slicing:** Highlights a specific range of colors in an image while background is set

to gray. A prototype color vector $\mathbf{a}$ and distance threshold D_0 are defined:

$$s_i = \begin{cases} r_i & \text{if } \|\mathbf{z} - \mathbf{a}\| \leq D_0 \\ 0.5 & \text{if } \|\mathbf{z} - \mathbf{a}\| > D_0 \end{cases} \quad (10)$$

- **Tone and Color Corrections:** Adjusting brightness, contrast, and gamma on color channels. Applying gamma correction on Intensity (I) in HSI space preserves the original hue and saturation, avoiding unwanted color shifts that occur when applying it directly on individual RGB channels.

3. Color Image Smoothing and Sharpening

3.1 Color Image Smoothing

Smoothing can be performed either in full-color vector space or by processing individual channels separately.

For a neighborhood S_{xy} , the average vector is:

$$\bar{\mathbf{c}}(x, y) = \frac{1}{K} \sum_{(s,t) \in S_{xy}} \mathbf{c}(s, t) = \begin{bmatrix} \frac{1}{K} \sum r_R(s, t) \\ \frac{1}{K} \sum r_G(s, t) \\ \frac{1}{K} \sum r_B(s, t) \end{bmatrix} \quad (11)$$

Since vector addition is performed component-wise, smoothing each RGB channel independently yields the exact same result as full-color vector smoothing.

3.2 Color Image Sharpening

Sharpening utilizes the Laplacian operator. In RGB vector space:

$$\nabla^2 \mathbf{c}(x, y) = \begin{bmatrix} \nabla^2 c_R(x, y) \\ \nabla^2 c_G(x, y) \\ \nabla^2 c_B(x, y) \end{bmatrix} \quad (12)$$

Like smoothing, color sharpening can be executed by applying the Laplacian operator independently to the Red, Green, and Blue channels.

4. Color Segmentation

Color segmentation partitions an image into regions based on color features.

4.1 Segmentation in RGB Space

Given a sample set of prototype pixels representing the target color, we compute their mean vector $\mathbf{a}$ and covariance matrix $\mathbf{C}$.

- **Euclidean Distance Classifier:** If color features are uncorrelated, we classify pixel $\mathbf{z}$ based on Euclidean distance:

$$D(\mathbf{z}, \mathbf{a}) = \|\mathbf{z} - \mathbf{a}\| = \sqrt{(z_R - a_R)^2 + (z_G - a_G)^2 + (z_B - a_B)^2} \quad (13)$$

A pixel is segmented if $D(\mathbf{z}, \mathbf{a}) \leq D_0$.

- **Mahalanobis Distance Classifier:** Accounts for correlations between channels:

$$D_M(\mathbf{z}, \mathbf{a}) = \sqrt{(\mathbf{z} - \mathbf{a})^T \mathbf{C}^{-1} (\mathbf{z} - \mathbf{a})} \quad (14)$$

4.2 Segmentation in HSI Space

Typically performed by thresholding Hue (H) and Saturation (S) channels:

- Intensity is often ignored, making the segmentation robust to illumination variations and shadows.

5. Worked Examples

RGB to HSI Conversion

Problem: Convert the normalized RGB pixel color vector $\mathbf{c} = [0.2, 0.8, 0.4]^T$ to HSI coordinates.

Solution: Given $R = 0.2$, $G = 0.8$, $B = 0.4$.

1. Calculate Intensity (I):

$$I = \frac{1}{3}(R + G + B) = \frac{1}{3}(0.2 + 0.8 + 0.4) = \frac{1.4}{3} \approx 0.467$$

2. Calculate Saturation (S): First, find $\min(R, G, B) = \min(0.2, 0.8, 0.4) = 0.2$.

$$S = 1 - \frac{3}{R + G + B} [\min(R, G, B)] = 1 - \frac{3}{1.4}(0.2) = 1 - \frac{0.6}{1.4} = 1 - 0.429 = 0.571$$

3. Calculate Hue (H): First, calculate the angle θ :

$$\begin{aligned} \text{Numerator} &= \frac{1}{2} [(R - G) + (R - B)] = \frac{1}{2} [(0.2 - 0.8) + (0.2 - 0.4)] \\ &= \frac{1}{2} [-0.6 - 0.2] = -0.4 \end{aligned}$$

$$\begin{aligned} \text{Denominator} &= \sqrt{(R - G)^2 + (R - B)(G - B)} = \sqrt{(0.2 - 0.8)^2 + (0.2 - 0.4)(0.8 - 0.4)} \\ &= \sqrt{(-0.6)^2 + (-0.2)(0.4)} = \sqrt{0.36 - 0.08} = \sqrt{0.28} \approx 0.529 \end{aligned}$$

Compute $\cos \theta$:

$$\cos \theta = \frac{-0.4}{0.529} \approx -0.756$$

Therefore:

$$\theta = \cos^{-1}(-0.756) \approx 139.1^\circ$$

Since $B = 0.4$ and $G = 0.8$, we have $B \leq G$. Thus, Hue is:

$$H = \theta = 139.1^\circ$$

Final Result: The HSI coordinate is:

$$H = 139.1^\circ, \quad S = 0.571, \quad I = 0.467$$

Quick Revision — Module III

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Key Formulas at a Glance

Intensity (HSI)	$I = \frac{1}{3}(R + G + B)$
Saturation (HSI)	$S = 1 - \frac{3}{R+G+B} [\min(R, G, B)]$
Luminance (YUV)	$Y = 0.299R + 0.587G + 0.114B$
RGB Color Complement	$s_R = 1 - r_R, \quad s_G = 1 - r_G, \quad s_B = 1 - r_B$
RGB Euclidean distance	$D(\mathbf{z}, \mathbf{a}) = \sqrt{(z_R - a_R)^2 + (z_G - a_G)^2 + (z_B - a_B)^2}$
Mahalanobis distance	$D_M(\mathbf{z}, \mathbf{a}) = \sqrt{(\mathbf{z} - \mathbf{a})^T \mathbf{C}^{-1} (\mathbf{z} - \mathbf{a})}$

One-Line Definitions

- **Additive Color Model:** Model (like RGB) where primary light sources are added together to create other colors.
- **Hue:** An attribute associated with the dominant wavelength in a mixture of light waves, representing pure color.
- **Saturation:** A measure of the relative purity of a color, indicating how much white light is mixed with it.
- **Color Slicing:** A transformation that isolates a sample color band in an image by zeroing out distant color vectors.
- **Full-Color Processing:** Processing pixels directly as multi-dimensional color vectors rather than scalar components.
- **Chrominance:** The color-carrying portion of a video signal, represented by U and V components in YUV.

Frequently Asked / Viva Questions

1. **Why is the HSI color model preferred over RGB for human interface design?**
Because HSI decouples intensity (brightness) from color features (hue/saturation), which mimics how the human brain perceives and describes colors.
2. **Which color model is used in video/television broadcasting? Why?**
YUV (or YC_bC_r), because it separates luminance (Y) from chrominance. This allows compatible broadcasting to black-and-white TVs and compression of color bandwidth without affecting perceived quality.
3. **Can we smooth a color image by filtering RGB channels independently?**
Yes. Since the average of vectors is equal to the vector containing the averages of their components, independent channel filtering produces mathematically identical results to full-color vector smoothing.

Frequently Asked / Viva Questions (contd.)

4. What is the complement of Red in RGB space?

Under normalized RGB space, Red is $(1, 0, 0)$. Its complement is $(1 - 1, 1 - 0, 1 - 0) = (0, 1, 1)$, which represents Cyan (a secondary color).

5. What is the main advantage of segmenting color images in HSI space?

By thresholding only Hue (H) and Saturation (S) and ignoring Intensity (I), the segmentation becomes robust to shadows and spatial variations in lighting.

6. Explain the physical meaning of a Saturation value of 0.

A saturation of 0 means the color is completely diluted with white light, representing a neutral shade of gray.

7. Why do we apply gamma correction to the Intensity channel in HSI rather than RGB channels?

Applying it to RGB channels independently alters their relative ratios, causing severe shifts in hue. Correcting the Intensity channel in HSI preserves hue and saturation.

8. What is the difference between Euclidean and Mahalanobis distance in RGB segmentation?

Euclidean distance assumes all color channels are independent and have equal variance. Mahalanobis distance uses the covariance matrix to account for channel correlations.

9. What are subtractive primary colors?

Cyan, Magenta, and Yellow. They are used in color printing and act as filters that absorb specific wavelengths of light.

10. What does the angle θ represent in HSI?

The angle θ represents the starting hue reference vector. It is measured counterclockwise starting from the Red axis (0°).

Comparison of Color Models

Attribute	RGB	YUV	HSI
Primary Focus	Hardware (displays)	Video coding/TV	Human perception
Components	Red, Green, Blue	Luminance, Chrominance	Hue, Saturation, Intensity
Decoupled Brightness	No	Yes (via Y)	Yes (via I)